

High resolution protein separations using affinity ultrafiltration with small charged ligands

Suma Rao, Andrew L. Zydney*

Department of Chemical Engineering, The Pennsylvania State University, University Park, PA 16802, USA

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Abstract

Although the feasibility of affinity ultrafiltration was demonstrated more than 20 years ago, commercial applications have not developed due to the high cost and practical limitations of the large macroligands needed for highly selective separations. The objective of this study was to examine the use of small charged affinity ligands for protein purification by exploiting electrostatic interactions between the charged complex and an electrically-charged membrane. Experiments were performed using bovine serum albumin and ovalbumin with Cibacron Blue as the affinity ligand. Negatively charged versions of a composite regenerated cellulose membrane were generated by covalent attachment of a sulfonic acid functionality. Binding experiments were used to identify appropriate conditions for protein separations. The selectivity for the separation of BSA and ovalbumin was a function of the solution conditions, Cibacron Blue concentration, and membrane charge, with the addition of Cibacron Blue causing a 30-fold increase in selectivity. A diafiltration process was performed at the optimal conditions, giving a BSA product with a purification factor of more than 90-fold and a yield greater than 90%. These results clearly demonstrate the potential of using a small charged affinity ligand for high resolution protein separations.

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1. Introduction

Affinity ultrafiltration can provide high resolution separations by exploiting the selectivity of biomolecular binding interactions while maintaining the high throughput characteristics of conventional ultrafiltration processes. Mattiasson and Ramstrop [1] demonstrated the feasibility of using affinity ultrafiltration for the purification of concanavalin A using heat killed *Saccharomyces cerevisiae* as the affinity ligand. The separation was performed using a standard hollow fiber membrane with a molecular weight cut-off of 1000 kD which was fully retentive to the very large binding complex formed between concanavalin A and the *S. cerevisiae* while allowing unbound species to pass relatively freely through the membrane.

Subsequent studies of affinity ultrafiltration have examined the isolation of urokinase using *N*-acryloyl-*m*-aminobenzamide

copolymerized with acrylamide as an affinity macroligand [2], the purification of avidin using biotinylated liposomes [3], the isolation of human serum albumin and lysozyme using Cibacron Blue bound to agarose [4], and the recovery of bovine serum albumin (BSA) using highly substituted Blue sephraose [5]. Affinity ultrafiltration has also been used to separate chiral molecules using stereoselective macroligands. For example, BSA has been used for the optical resolution of racemic mixtures of tryptophan [6,7] and human serum albumin has been used for the separation of ibuprofen [8].

These applications of affinity ultrafiltration all used very large macroligands, relative to the size of the desired product, to obtain the desired retention of the ligand-product complex using conventional ultrafiltration membranes. However, these macroligands tend to be very expensive, which has severely limited the viability of affinity ultrafiltration for commercially relevant separation processes. In addition, the use of small ligands bound to large carriers, e.g., Cibacron Blue bound to agarose [4], can result in a loss of ligand accessibility, mass transfer limitations, and/or problems with leaching of the ligand from the large carrier over time. The net result is that there are currently no large-scale

* Corresponding author at: Department of Chemical Engineering, The Pennsylvania State University, University Park, PA 16802, USA.
Tel.: +814 863 7113; fax: +814 865 7846.

E-mail address: zydney@engr.psu.edu (A.L. Zydney).

commercial applications of affinity ultrafiltration for the purification of high value products.

Affinity ultrafiltration would be much more attractive commercially if one could employ a relatively small inexpensive affinity ligand to effect the separation. Recent work by Rao and Zydney [9] has demonstrated that small charged dyes can dramatically alter the rate of protein transport through electrically-charged ultrafiltration membranes due to the strong electrostatic interactions between the dye–protein complex and the charged membrane. For example, the addition of 1 g/L of Cibacron Blue to an 8 g/L solution of bovine serum albumin reduced the BSA sieving coefficient by more than two orders of magnitude.

The objective of this study was to demonstrate the feasibility of using affinity ultrafiltration with the small charged ligand Cibacron Blue for separation of the model protein mixture of BSA and ovalbumin. Ovalbumin and BSA have similar size and surface charge, but previous data suggest that ovalbumin has limited binding interactions with Cibacron Blue [10]. Experiments were performed to identify the optimal conditions for the separation of ovalbumin and BSA using this affinity ultrafiltration process. The results clearly demonstrate that high resolution protein separations can be achieved using a small charged affinity ligand with high binding specificity in combination with an electrically-charged ultrafiltration membrane to retain the like-charged affinity complex.

2. Material and methods

2.1. Protein solutions

Experiments were performed using bovine serum albumin (>96%, essentially fatty acid free, catalogue # A-6003 from Sigma Chemical, St. Louis, MO). BSA has an isoelectric point (pI) of approximately 4.9 and a molecular weight of 67 kD. Ovalbumin (catalogue # A-5503, Grade V, minimum 98%, agarose gel electrophoresis, from Sigma Chemical, St. Louis, MO) has an isoelectric point (pI) of approximately 4.5 and a molecular weight of 44 kD.

Cibacron Blue 3GA (Catalogue # C-9534, Sigma Chemical) was used as the small affinity ligand. Cibacron Blue has a molecular weight of 0.774 kD and contains three sulfonic acid groups attached to an aromatic ring structure [11]. Previous work by Rao and Zydney [9] indicated that as many as 10 molecules of Cibacron Blue can bind to a single molecule of BSA, probably through the fatty acid and anion binding sites on the protein.

Buffer solutions were prepared by dissolving pre-weighed amounts of the appropriate salts in deionized water obtained from a NANOpure Diamond water purification system (Barnstead Thermolyne Corporation, Dubuque, IA) with a resistivity greater than 18 M Ω -cm. An acetate buffer composed of CH₃COONa and CH₃COOH (EM Science, Gibbstown, NJ) was used for experiments at pH 5.0. The solution pH was measured using a 420APlus pH meter (Thermo Orion, Beverly, MA). The ionic strength was adjusted by adding NaCl (Mallinckrodt Baker, Inc., Phillipsburg, NJ), with the solution conductivity measured using a 105A plus conductivity meter (Thermo Orion, Beverly, MA). All buffer solutions were prefiltered through 0.2 μ m

pore size Supor 200 membranes (Pall Corp., Ann Arbor, MI) to remove particulates and undissolved salts.

Protein solutions were prepared by slowly dissolving the protein powder in the desired buffer, with the resulting solution filtered through a 0.22 μ m syringe filter (Costar Corp., Cambridge, MA) to remove any protein aggregates immediately prior to use. Protein solutions were used within 24 h to minimize the likelihood of protein aggregation or denaturation during storage.

2.2. Assays

Concentrations of the individual proteins in the absence of the dye were determined spectrophotometrically with a UV–vis spectrophotometer (UV mini 1240, Shimadzu, Kyoto, Japan) using the natural absorbance at 280 nm. Actual concentrations were evaluated by comparison of the measured absorbance with that of standard solutions of known concentrations. Cibacron Blue concentrations were also determined spectrophotometrically using the natural absorbance at 616 nm. The presence of protein had no effect on the absorbance of the dye at 616 nm over the range of concentrations examined in this study.

BSA concentrations in the presence of Cibacron Blue were evaluated by capillary zone electrophoresis (Model G1600A, Agilent Technologies, Palo Alto, CA) since the strong color of Cibacron Blue interfered with the absorbance measurements. Capillary electrophoresis was performed using fused silica capillaries with an inner diameter of 50 μ m and an effective length of 72 cm (Catalogue # G 1600-62211, Agilent Technologies, Palo Alto, CA). The applied voltage was maintained at 25 kV, and a 20 mM borate buffer at pH 9.3 (Catalogue # 8500-6782, Agilent Technologies, Waldbronn, Germany) was used as the running buffer. Protein detection was at a wavelength of 195 nm, with the protein concentration determined by numerical integration of the area under the corresponding peak. Actual concentrations were evaluated using previously constructed calibration curves determined for the protein in the presence of Cibacron Blue.

Protein concentrations in the binary mixture were analyzed by size exclusion chromatography (Agilent 1100 series quaternary HPLC system). Assays were performed using a Superdex 75 column with 13 μ m particle size and 10⁵ MW exclusion limit (GE Healthcare, Piscataway, NJ). Column calibration was done using binary protein solutions of known concentration. The column was first equilibrated with fresh buffer at a flow rate of approximately 0.3 mL/min for 180 min. This also served to flush both the sample and reference cells in the refractive index detector (Agilent 1100 series). Column equilibration was confirmed by tracking the baseline refractive index (RI). The mobile phase was a 50 mM phosphate buffer with 0.15 M NaCl at a flow rate of 0.2 mL/min. Protein samples (50 μ l) were injected by an autosampler with the data analyzed using Agilent ChemStation software on a Dell Celeron Computer.

2.3. Membrane preparation

All filtration experiments were performed using composite regenerated cellulose (CRC) membranes obtained from Millipore Corp. (Bedford, MA) with nominal molecular weight

cut offs of 30 kD or 100 kD. These cellulosic membranes are nearly uncharged and have very low protein adsorption due to their high degree of hydrophilicity. Individual membrane discs (25 mm diameter) were cut from large flat sheets using a specially designed cutting device.

A negatively-charged version of the 100 kD membrane was made in our laboratory by the covalent attachment of negatively charged sulfonic acid groups to the surface of the membrane using the base-activated chemistry developed by van Reis [12]. Membranes were first flushed with deionized distilled water to remove any residual storage agents. The membranes were then equilibrated with 0.1 N NaOH and immersed in a 2 M solution of 3-bromopropanesulfonic acid sodium salt (Catalogue #B2912, Sigma Chemical) in 0.1 N NaOH for fixed periods of time ranging from 2 to 24 h to vary the extent of surface modification. The membranes were then flushed with approximately 100 L/m² of distilled water followed by storage in 0.1 N NaOH.

2.4. Equilibrium binding experiments

Binding interactions were evaluated using a stirred ultrafiltration cell which allowed protein-free samples to be collected through the UF membrane. A protein–dye mixture was added to a 25 mm diameter stirred UF cell (Model 8010, Amicon Corp., Beverly, MA) which contained either a 30 kD nominal molecular weight cut-off unmodified CRC membrane (for experiments with BSA) or a 10 kD membrane (for ovalbumin). The stirrer was set at 600 rpm using a Strobotac Type 1531-AB strobe light (General Radio Co., Concord, MA). The device was then air-pressurized to $\Delta P = 21$ kPa (3 psig), and small samples were periodically withdrawn through the membrane for off-line analysis of the free Cibacron Blue concentration. Data were also obtained with pure Cibacron Blue (in the absence of protein) to evaluate the sieving coefficient of free Cibacron Blue through the 30 and 10 kD membranes.

2.5. Protein filtration

Protein filtration experiments were also conducted in the 25 mm diameter stirred ultrafiltration cell. The stirred cell was initially filled with buffer taking care to remove all trapped air bubbles from the cell and tubing. The stirrer speed was set to 600 rpm using a Strobotac Type 1531-AB strobe light, and the device was air-pressurized to the desired value. The membrane hydraulic permeability (L_p) was evaluated from the slope of data for the filtration velocity (J_v) as a function of the transmembrane pressure drop (ΔP):

$$L_p = \frac{J_v \mu}{\Delta P} \quad (1)$$

where μ is the solution viscosity. The stirred cell was then emptied and refilled with a protein – Cibacron Blue solution at the desired ionic strength and pH. The stirrer was reset, the cell was air-pressurized, and the filtrate flux was determined by timed collection using a digital balance (Model AG104, Mettler Toledo, Columbus, OH) with an accuracy of 0.1 mg. A minimum of 500 μ L of filtrate was flushed through the membrane to ensure

washout of the dead volume beneath the membrane at the bottom of the stirred cell. A 200 μ L filtrate sample was then collected for subsequent analysis of the protein and Cibacron Blue concentrations. The filtrate port was immediately clamped and a small (approximately 200 μ L) sample of the bulk solution was obtained directly from the stirred cell. The stirred cell was then emptied, rinsed, and refilled with buffer, with the hydraulic permeability of the used membrane evaluated to provide a measure of the extent of fouling. All experiments were performed at room temperature (22 ± 3 °C) using the same lots of protein.

2.6. Diafiltration

The actual separation of ovalbumin and BSA was performed using the Amicon stirred cell operated in diafiltration mode. The stirred cell was initially filled with 10 ml of the protein mixture. A separate reservoir was filled with a buffer solution (without protein or dye) and attached to the stirred cell with silicone tubing. The filtrate velocity was maintained at nearly a constant value over the course of the diafiltration by a peristaltic pump connected to the filtrate line. The filtrate flux created a vacuum in the stirred cell causing fresh buffer from the reservoir to be drawn into the cell from the solution reservoir at a rate equal to the volumetric filtration rate. Filtrate samples were collected throughout the experiment with periodic samples taken directly from the stirred cell to evaluate the protein concentrations in the retentate.

2.7. Membrane charge

The membrane charge was evaluated from streaming potential measurements following the procedure described by Burns and Zydney [13]. The membrane was sealed between two Plexiglas chambers which were then filled with a 10 mM NaCl solution. The solution conductivity was measured using a 105A plus conductivity meter (Thermo Orion, Beverly, MA). After carefully removing any entrapped air bubbles, the system was pressurized by adjusting the height of the solution (feed) reservoir. The system was allowed to stabilize for approximately 12 h at which point the transmembrane voltage (E_z) was measured using a high impedance 8060A True RMS Multimeter (Fluke Corporation, Everett, WA) connected to the two electrodes. Data were obtained at several pressures in order to evaluate the streaming potential.

The effective membrane surface charge density was evaluated from the slope of the voltage as a function of pressure from the Helmholtz–Smoluchowski equation [14]:

$$\frac{dE_z}{dP} = \left(\frac{\varepsilon_0 \varepsilon_r \zeta}{\mu \Lambda} \right) \quad (2)$$

where ζ is the zeta potential, Λ the solution conductivity, ε_0 the electrical permittivity of a vacuum, and ε_r the dielectric constant of the fluid. The effective surface charge density of the membrane can be calculated directly from the zeta potential as:

$$\sigma_p = 4c_0 F k^{-1} \sinh \left(\frac{F \zeta}{2RT} \right) \quad (3)$$

where c_0 is the bulk salt concentration, F is the Faraday's constant, R is the gas constant, T is the absolute temperature, and κ^{-1} is the thickness of the electrical double layer

$$\kappa^{-1} = \left[\frac{F^2}{\varepsilon_0 \varepsilon_r RT} \sum z_i^2 C_i \right]^{1/2} \quad (4)$$

where z_i and C_i are the valence and concentration of each ion. Eqs. (2) and (3) neglect the effects of surface conductance and a thick double layer, thus the calculated values of σ_p represent an effective surface charge density of the membrane pores.

3. Results and analysis

3.1. Binding isotherms

Before examining the effects of Cibacron Blue on protein separations, we first evaluated the extent of Cibacron Blue binding to BSA and ovalbumin individually over a range of conditions using the approach described by Rao and Zydney [9]. The amount of bound Cibacron Blue was determined from a simple mass balance:

$$C_{\text{bound}} = C_{\text{total}} - C_{\text{free}} \quad (5)$$

where the total concentration of Cibacron Blue in the solution was evaluated directly from the mass of dye added to a known volume of buffer. The concentration of free Cibacron Blue was evaluated directly from the measured filtrate concentration as:

$$C_{\text{free}} = \frac{C_{\text{filtrate}}}{S_0} \quad (6)$$

where the observed sieving coefficient for free Cibacron Blue (S_0), defined as the ratio of the solute concentration in the filtrate to that in the bulk solution, was evaluated immediately prior to performing the binding experiment using the same membrane and the same concentration of pure Cibacron Blue that was added to the protein solution. Typical sieving coefficients for Cibacron Blue were greater than 0.5 for both the 10 and the 30 kD CRC membrane. The small amount of Cibacron Blue retention is consistent with previous studies of the transport of small charged solutes through charged ultrafiltration membranes [15]. There was no evidence of leakage of the 67 kD BSA through the 30 kD membrane, or of the 44 kD ovalbumin through the 10 kD membrane, as determined by capillary electrophoresis of the filtrate samples.

Typical binding data for BSA and ovalbumin at pH 5 and 10 mM ionic strength are shown in Fig. 1. These data were obtained from two separate sets of experiments, one using a solution containing 5 g/L BSA (top panel) and one performed with a solution containing 15 g/L ovalbumin (bottom panel). The results are plotted as the molar concentration of bound Cibacron Blue normalized by the molar protein concentration as a function of the concentration of free Cibacron Blue. The extent of binding in the case of BSA increases rapidly at low Cibacron Blue concentrations, but then levels off at high concentrations. The binding to BSA (5 g/L) is saturated above a Cibacron Blue concentration of about 0.15 g/L while the binding to ovalbumin

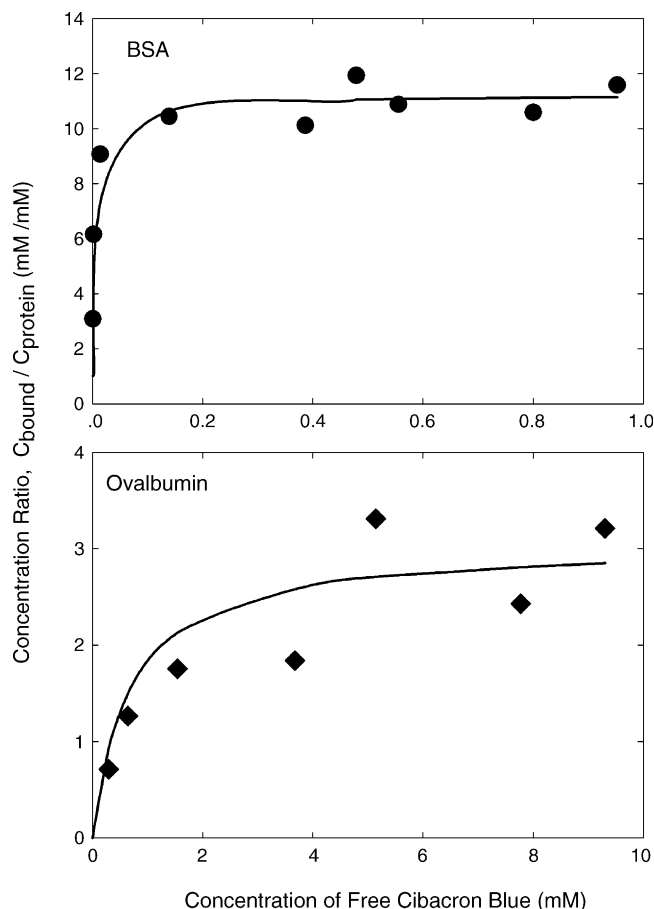


Fig. 1. Binding isotherms for Cibacron Blue at pH 5.0 in a 10 mM acetate buffer at a BSA concentration of 5 g/L (top panel) and an ovalbumin concentration of 15 g/L (bottom panel).

(15 g/L) approaches the saturation value only above a Cibacron Blue concentration of about 4 g/L, indicating much stronger binding to BSA. In addition, the saturation value for BSA corresponds to approximately 11 moles of bound Cibacron Blue per mole of BSA compared to only 2.7 moles of bound Cibacron Blue per mole of ovalbumin. The BSA data are consistent with previous studies which have clearly demonstrated that Cibacron Blue can bind to multiple fatty acid binding sites on the surface of BSA [16,17].

Experiments performed over a range of pH showed a sharp reduction in the amount of Cibacron Blue binding per mole of BSA at higher pH, which is probably due to the increased electrostatic repulsion between the more negatively-charged protein and the negatively-charged dye. The best fit values of n , the number of moles of Cibacron Blue bound per mole of protein, under different conditions are summarized in Table 1. The greatest amount of Cibacron Blue binding to BSA was obtained at pH 5 with $n = 11$, with this value decreasing to less than $n = 3$ at pH 8. The maximum degree of selectivity for Cibacron Blue binding to BSA versus ovalbumin was obtained around pH 5 and at relatively low Cibacron Blue concentrations. For example, at pH 5 most of the binding sites on BSA were saturated at a Cibacron Blue concentration of 12 moles of Cibacron Blue per mole of BSA. The amount of Cibacron Blue bound to ovalbumin under

Table 1
Binding interactions with Cibacron Blue

Protein	pH, Ionic strength 10 mM	Moles Cibacron Blue/mole protein (<i>n</i>)
BSA	5.0	11.0
BSA	6.0	5.7
BSA	7.0	4.2
BSA	8.0	2.3
Ovalbumin	5.0	2.7

these conditions was only about 0.5 moles of Cibacron Blue per mole of ovalbumin based on a linear extrapolation of the data at the lowest dye concentrations in Fig. 1.

3.2. Protein sieving

Fig. 2 shows experimental data for the protein sieving coefficients, evaluated separately for solutions of either BSA or ovalbumin, both with and without the addition of Cibacron Blue as an affinity ligand. Experiments were performed using a series of 100 kD CRC membranes, each produced by chemical modification of a separate membrane disk for different reaction times up to 24 h. The surface charge density of the resulting membranes was evaluated using streaming potential measurements,

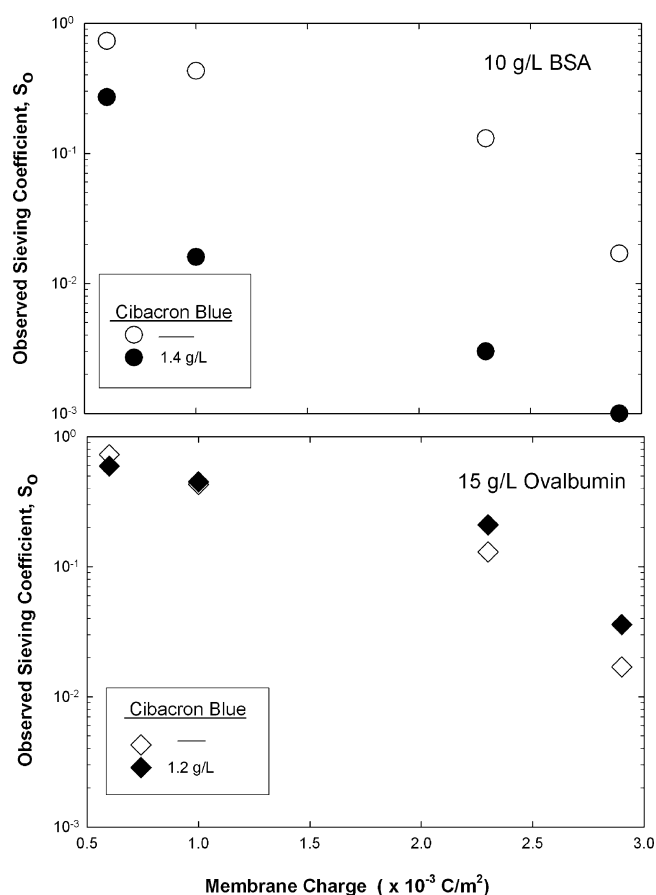


Fig. 2. Observed protein sieving coefficients in a pH 5, 10 mM acetate buffer as a function of membrane surface charge. Data are shown in the absence (open symbols) and presence (filled symbols) of Cibacron Blue.

with values ranging from -0.0005 C/m^2 for the unmodified CRC membrane to -0.0029 C/m^2 for the membrane exposed to the charging ligand for 24 h. Note that these are only rough estimates of the surface charge density since no attempt was made to account for the effects of double layer overlap or surface conduction in the analysis of the streaming potential data [14].

The top panel shows results for a 10 g/L solution of BSA in a pH 5, 10 mM acetate buffer both alone and with a 12:1 molar ratio of Cibacron Blue per mole of BSA (corresponding to a 1.4 g/L Cibacron Blue concentration). The BSA sieving coefficient in the absence of Cibacron Blue decreases from a value of 0.60 for the unmodified CRC membrane to a value of 0.02 for the membrane with a surface charge density of -0.0029 C/m^2 due to the electrostatic interactions between the weakly charged BSA and the sulfonic acid groups on the highly charged membrane. This behavior is discussed in more detail by Mehta and Zydney [18]. The reduction in the BSA sieving coefficient is even more pronounced in the presence of Cibacron Blue, with the protein sieving coefficient decreasing by more than two orders of magnitude over this same range of membrane charge. This behavior is a direct result of the strong degree of Cibacron Blue binding to BSA, which significantly increases the net negative charge of the protein thereby increasing the electrostatic exclusion from the negatively-charged membrane. Model calculations performed by Rao and Zydney [9] under similar conditions indicate that the net charge of BSA increases by approximately 14 negative charge units upon the binding of 11 moles of Cibacron Blue per mole of BSA due to the effects of charge regulation.

The corresponding results for the ovalbumin sieving coefficient are shown in the bottom panel of Fig. 2 for experiments performed using a 15 g/L ovalbumin solution. In the absence of Cibacron Blue, the ovalbumin sieving coefficient decreases from 0.73 for the unmodified membrane to 0.018 for the membrane with a surface charge density of -0.0029 C/m^2 . In contrast to the data for BSA, the addition of 1.2 g/L of Cibacron Blue has relatively little effect on ovalbumin sieving, with the sieving coefficients in the presence of Cibacron Blue being slightly larger than those obtained for the pure protein, particularly for the more highly charged membranes. The small increase in the ovalbumin sieving coefficient is a direct result of the low degree of Cibacron Blue binding to ovalbumin under the conditions of these experiments. The addition of Cibacron Blue has little effect on the ovalbumin charge, but it increases the ionic strength of the solution due to the presence of the free (unbound) dye, slightly shielding the electrostatic interactions with charged membranes.

In the absence of Cibacron Blue, the transmission of ovalbumin is slightly larger than that of BSA, reflecting the somewhat smaller size of the ovalbumin (Stokes–Einstein radius of 2.8 nm compared to 3.5 nm for BSA). The use of the more highly charged membranes had no significant effect on the selectivity between the two proteins (defined as the ratio of the protein sieving coefficients), due to the very similar surface charge characteristics of BSA and ovalbumin at pH 5. The addition of Cibacron Blue causes a large difference in the protein sieving coefficients, with the sieving coefficient of BSA decreasing sharply (particularly for the more highly charged membranes) while the sieving coefficient of ovalbumin increases slightly. The

net result is a dramatic increase in the selectivity between the two proteins.

The best membrane for the separation of ovalbumin and BSA can be determined using the process optimization diagrams developed by van Reis and Saksena [19], with the separation performance determined by two key parameters: the selectivity $\psi = S_{\text{ovalbumin}}/S_{\text{BSA}}$ and the throughput $J_v \Delta S = J_v(S_{\text{ovalbumin}} - S_{\text{BSA}})$. The selectivity and throughput for the separation of BSA and ovalbumin were estimated from the data in Fig. 2 assuming that the behavior for the binary mixture will be similar to that obtained for the separate experiments using BSA and ovalbumin. The estimated selectivity for the membrane with the highest surface charge density of -0.0029 C/m^2 was quite high ($\psi = 36$) due to the very strong retention of the BSA–dye complex. However, this membrane had a relatively low throughput ($J_v \Delta S = 3.5 \mu\text{m/s}$) due to the fairly high retention of the ovalbumin. In contrast, the membrane with a surface charge density of -0.0011 C/m^2 has a higher throughput ($J_v \Delta S = 43 \mu\text{m/s}$) but a lower selectivity ($\psi = 28$). The best separation characteristics, as determined from the optimization diagrams presented by van Reis and Saksena [19], were predicted for the membrane having a surface charge density of -0.0023 C/m^2 and hence this membrane was used in all subsequent experiments.

Fig. 3 shows the effect of the Cibacron Blue concentration on the protein sieving coefficients for the negatively-charged 100 kD CRC membrane. These experiments were performed with a binary mixture containing 10 g/L BSA and 6 g/L ovalbumin in a pH 5.0, 10 mM acetate buffer. The BSA sieving coefficient initially decreases upon addition of Cibacron Blue, attaining a minimum value ($S_0 = 0.0064$) at a Cibacron Blue concentration of approximately 1.4 g/L corresponding to the addition of 12 moles of dye per mole of BSA. Higher concentrations of Cibacron Blue cause an increase in BSA transmission, with the observed sieving coefficient at high ligand concentrations approaching a value of 0.04. This increase in BSA transmission is due to the shielding of the electrostatic interactions associated with the presence of significant amounts of free (unbound)

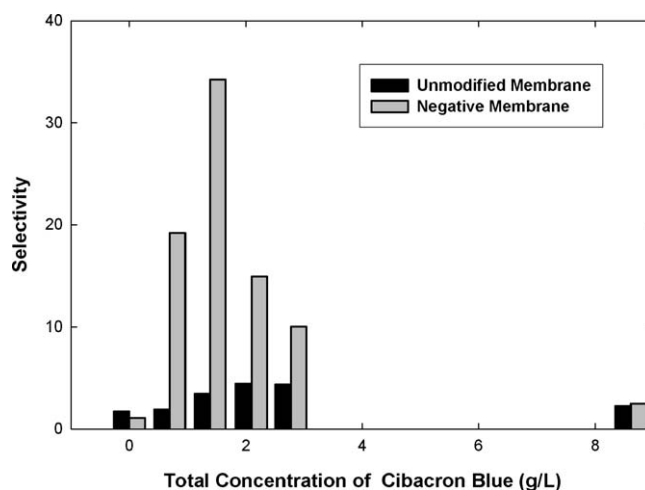


Fig. 4. Selectivity for the separation of BSA and ovalbumin as a function of the total Cibacron Blue concentration for an unmodified and negatively-charged membrane.

Cibacron Blue in the solution, an effect that was discussed in some detail by Rao and Zydney [9]. In contrast to the data for BSA, the observed sieving coefficient of ovalbumin increases with increasing Cibacron Blue concentration even at very low ligand concentrations, but then passes through a weak maximum at a Cibacron Blue concentration of about 2.1 g/L. The reduction in ovalbumin transmission at high Cibacron Blue concentrations could be due to some ligand binding under these conditions, consistent with the high concentrations required for saturation of the ovalbumin binding sites seen in Fig. 1. It is also possible that there is some Cibacron Blue binding to the membrane at very high ligand concentrations, which would lead to an increase in the net negative charge of the membrane and a corresponding reduction in protein transmission.

The experimental data in Fig. 3 have been replotted in Fig. 4 in terms of the selectivity, which is defined as the ratio of the transmission of ovalbumin to that of BSA ($S_{\text{OVA}}/S_{\text{BSA}}$). Also shown for comparison are results obtained with a neutral (un-modified) CRC membrane. The selectivity in the absence of Cibacron Blue is negligible for both membranes, with a value less than 1.1. The selectivity for the negatively-charged membrane increases dramatically upon addition of Cibacron Blue, attaining a maximum value of more than 30-fold at a Cibacron Blue concentration of 1.4 g/L, which corresponds to the minimum in the BSA sieving coefficient seen in Fig. 3. Further increases in the Cibacron Blue concentration cause a reduction in the selectivity, with the selectivity decreasing to below three in the presence of 8.6 g/L of Cibacron Blue. This decrease in selectivity is due to the reduction in ovalbumin transmission coupled with the increase in BSA transmission seen at high concentrations of Cibacron Blue, both of which are due to the shielding of the electrostatic interactions by the excess (unbound) Cibacron Blue. It is important to note that the maximum selectivity with the unmodified CRC membrane is less than 5.0 due to the absence of any significant electrostatic interactions with the unmodified membrane.

All of the data in Figs. 1–4 were obtained using an acetate buffer with 10 mM ionic strength. The effect of the buffer ionic

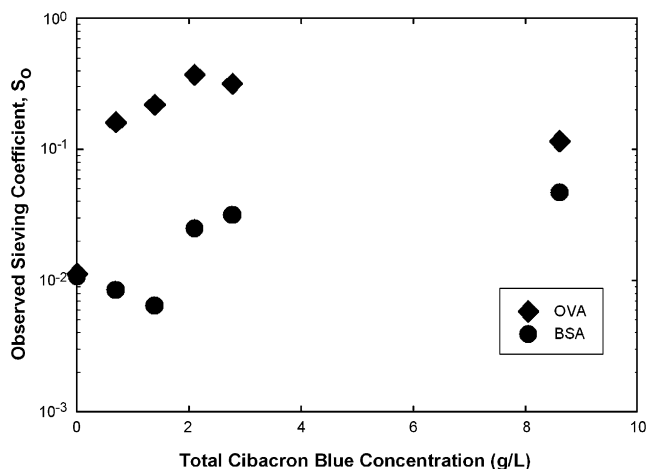


Fig. 3. Observed sieving coefficients for BSA and ovalbumin in a binary mixture for the negatively charged 100 kD CRC membrane. Data were obtained using a 10 mM acetate buffer at pH 5.0 over a range of Cibacron Blue concentrations.

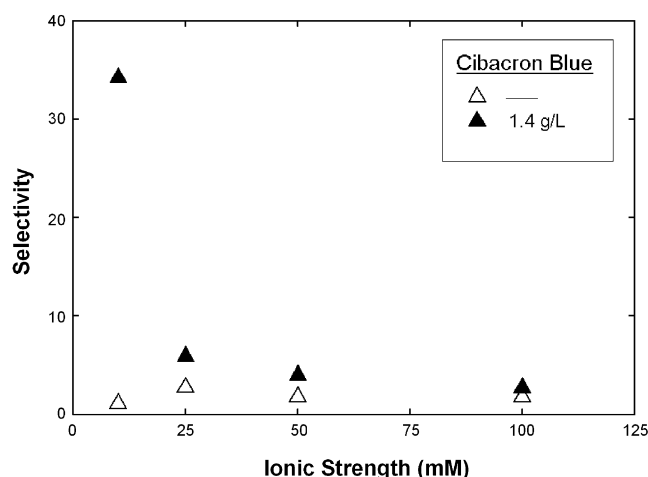


Fig. 5. Selectivity for the separation of BSA and ovalbumin as a function of the solution ionic strength both with and without Cibacron Blue.

strength on the selectivity for the separation of BSA and ovalbumin is examined in Fig. 5. Experiments were performed using a binary mixture containing 10 g/L BSA and 6 g/L ovalbumin, both with and without Cibacron Blue at a concentration of 1.4 g/L. The ionic strength was adjusted by addition of NaCl to the 10 mM acetate buffer. The selectivity in the absence of Cibacron Blue increases from a value of 1 in the 10 mM ionic strength buffer to approximately 2.7 at an ionic strength of 25 mM before dropping to below 2 at higher ionic strength. The weak variation of the selectivity with ionic strength arises from the slightly different electrostatic interactions for BSA and ovalbumin. The sieving coefficients for both proteins increase with increasing ionic strength due to the shielding of the electrostatic interactions, but this effect is slightly larger for ovalbumin at the lower ionic strengths, with the reverse behavior seen at high ionic strengths, resulting in the small maximum in selectivity seen in Fig. 5. The addition of Cibacron Blue causes a dramatic increase in selectivity, particularly for the lowest ionic strength solution. The selectivity decreases with increasing ionic strength due to the shielding of the electrostatic interactions, with the selectivity at the highest ionic strength being nearly identical with and without Cibacron Blue.

3.3. Protein diafiltration

Based on the results in Figs. 4 and 5, a diafiltration process was designed for the actual separation of BSA and ovalbumin using a pH 5.0, 10 mM ionic strength acetate buffer with a negatively charged 100 kD CRC membrane having a nominal surface charge density of -0.0023 C/m^2 . The feed solution contained 2 g/L BSA and 1 g/L ovalbumin, with Cibacron Blue added at a concentration of 0.28 g/L corresponding to 12 moles of Cibacron Blue per mole of BSA. A control experiment was performed without Cibacron Blue to quantify the effects of the protein–ligand interactions. A small amount of NaCl was added to the buffer used in the control experiment so that the solution ionic strength was identical to that measured for the diafiltration in the presence of Cibacron Blue.

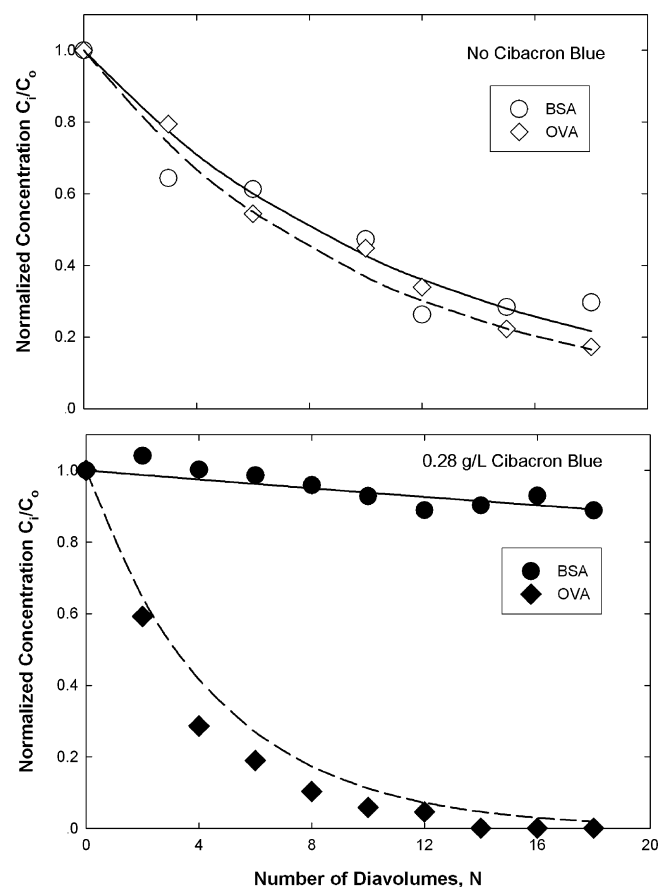


Fig. 6. Normalized protein concentrations during diafiltration experiments performed in the absence (top panel) and in the presence (bottom panel) of Cibacron Blue using a negatively charged 100 kD CRC membrane. Solid curves are model calculations as described in the text.

Fig. 6 shows data for the normalized protein concentration in the stirred cell as a function of the number of diavolumes (N), which is equal to the total collected filtrate volume divided by the constant volume in the stirred cell. Results are shown in the presence (bottom panel) and in the absence (top panel) of Cibacron Blue, with the solid curves representing the calculated values of the protein concentration evaluated by integration of the differential mass balance assuming a constant observed sieving coefficient [19]:

$$C/C_0 = \exp(-NS_0) \quad (7)$$

The observed sieving coefficients for the BSA and ovalbumin in the presence of Cibacron Blue were evaluated from the data in Figs. 4 and 5. The concentration of ovalbumin decreased rapidly for the diafiltration performed in the presence of Cibacron Blue, dropping to nearly undetectable values after 16 diavolumes. There was a small loss of BSA over the course of the diafiltration, with the BSA concentration decreasing from 2.0 to 1.75 g/L after 14 diavolumes. The diafiltration thus yielded a BSA product with a purification factor greater than 90-fold at a yield of more than 90%. Ovalbumin was collected in the permeate solution with a purification factor greater than 10-fold and essentially 100% yield.

In contrast to the results obtained in the presence of Cibacron Blue, the concentration of BSA decreased by approximately 75% for the diafiltration performed in the absence of the affinity ligand while the ovalbumin concentration decreased by 85%. The net result is a purification factor less than 2-fold for both BSA and ovalbumin, compared to the more than 90-fold purification for BSA seen in the presence of the Cibacron Blue. The solid and dashed curves in the top panel of Fig. 6 were determined by fitting the concentration data to Eq. (7), yielding S_0 for BSA of 0.085 and S_0 for ovalbumin of 0.1. The differences between these best fit values of the observed sieving coefficients and the data reported previously in Figs. 4 and 5 are likely due to the addition of NaCl to adjust the ionic strength of the acetate buffer, although there is also some inherent variability between experiments performed with different membrane samples.

Fig. 7 shows size exclusion chromatograms for the feed (top panel) and the final retentate solution (bottom panel) for the 16-diavolume diafiltration performed in the presence of the Cibacron Blue. The small peak at a retention time of 62.5 min is due to the presence of a small amount of BSA dimer in the feed, corresponding to approximately 13% of the main BSA peak at a retention time of 71 min. The ovalbumin appears as the large peak at 76 min; this peak is almost completely absent for the chromatogram at the end of the diafiltration, consistent

with the very high degree of BSA purification obtained in this experiment.

4. Discussion

Although a number of previous studies have shown the feasibility of using affinity ultrafiltration with large macroligands, the results presented in this study provide the first demonstration that it is possible to use small charged ligands for highly selective affinity ultrafiltration, with the product of interest retained by electrostatic interactions using an electrically-charged membrane. A selectivity of more than 30-fold was obtained for the separation of ovalbumin and BSA upon the addition of Cibacron Blue, compared to less than a two-fold selectivity in the absence of the affinity ligand. This dramatic increase in selectivity was a direct result of the strong electrostatic exclusion of the highly charged complex formed between BSA and the Cibacron Blue, with independent data for the binding interactions indicating that each molecule of BSA can bind as many as 10 molecules of Cibacron Blue.

The design of an effective affinity ultrafiltration process using a charged microligand is a complex function of the membrane charge, the solution pH and ionic strength, and the concentration of the affinity ligand. Low concentrations of the affinity ligand lead to inadequate selectivity due to the absence of significant binding while very high concentrations of the ligand reduce the electrostatic interactions due to the increase in ionic strength associated with the presence of a significant amount of free ligand in solution. The net result is the presence of an optimal ligand concentration, properly balancing the selectivity of the binding interactions with the strong electrostatic repulsion that occurs at low ionic strength. The membrane charge also has a large effect on the separation efficiency, with weakly charged membranes providing inadequate electrostatic repulsion.

An affinity diafiltration process was designed to obtain both BSA and ovalbumin as highly purified products in the retentate and filtrate, respectively. Appropriate conditions for the diafiltration were determined from the estimated values of selectivity and throughput. The resulting process gave a purification factor for BSA of greater than 90-fold at more than 90% yield after 14 diaolumes. The ovalbumin product was collected in the filtrate with a purification factor of more than 10-fold at nearly 100% yield. Although not shown, we were also able to recover the Cibacron Blue at the end of the process by performing a second diafiltration using a 10 kD membrane at high pH and in the presence of a chaotrope like NaSCN, conditions that have been shown previously to reverse the binding of Cibacron Blue to BSA [20]. This second diafiltration removed more than 99% of the Cibacron Blue after 8 diaolumes, with the purified BSA recovered in the retentate. Additional details on this process will be presented in a subsequent publication. Future studies will be required to examine the overall economics of this type of affinity ultrafiltration using a small charged microligand; however, the very low cost of the affinity dye compared to that of the large affinity macroligands used previously make this a potentially attractive approach for protein separations using affinity ultrafiltration.

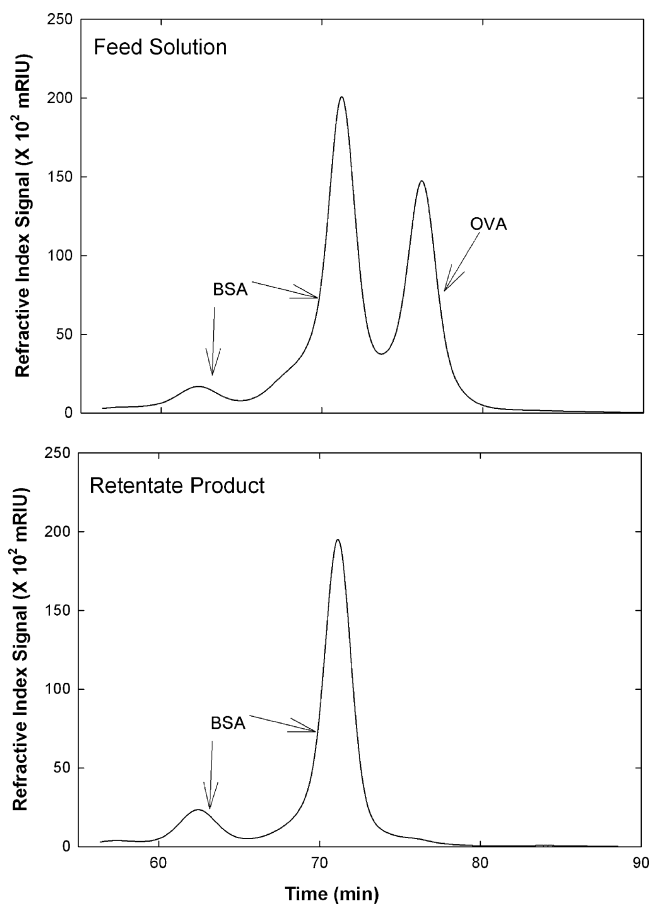


Fig. 7. Size exclusion chromatograms for the initial feed (top panel) and final retentate (bottom panel) for the diafiltration process performed in the presence of Cibacron Blue.

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